

Program: Diploma in Fire Technology and Safety	
Course Code: 4464	Course Title: Safety in Engineering Industries
Semester: 4	Credits: 0
Course Category: Program Core	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To understand the principles of safety in the process design and the importance of inherently safer design.
- To learn about pressure system design, including vessels, piping, valves, and pressure relief devices.
- To explore the stages of plant commissioning, inspection, and performance monitoring.
- To gain knowledge in plant operations, including emergency procedures, operational hazards, and equipment safety.
- To study the safety measures for storage systems, including those for hazardous chemicals and petroleum products.

Course Pre requisites:

Topic	Course code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the process design principles, inherently safer design in chemical reactors, and pressure system design	11	Understanding
CO2	Learn to perform plant commissioning, inspection, and monitor performance for various plant systems.	9	Applying
CO3	Gain practical knowledge in plant operations, including emergency procedures and safety protocols.	9	Applying

CO4	Understand plant maintenance, modification, emergency planning, and storage safety measures for hazardous materials.	14	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	3	3					
CO4	3	2					

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the process design principles, inherently safer design in chemical reactors, and pressure system design		
M1.01	Design process, conceptual design, and detail design.	2	Understanding
M1.02	Inherently safer design for chemical reactors.	1	Understanding
M1.03	Types of batch reactors, reaction hazard evaluation and reactor safety.	2	Applying
M1.04	Operating conditions and unit operations safety.	2	Understanding
M1.05	Pressure system design and pressure vessel design, standards and codes.	1	Understanding
M1.06	Overpressure protection, relief devices design, and flare/vent systems.	1	Applying
M1.07	Failures in pressure systems and their prevention.	2	Applying

Contents: Conceptual design, inherently safer design in reactors. Types of reactors: batch reactors and reaction hazard evaluation. Pressure vessels, pipes, valves, heat exchangers, and overpressure protection. Pressure relief devices, vacuum and thermal relief. Special situations and failures in pressure systems.			
CO2	Learn to perform plant commissioning, inspection, and monitor performance for various plant systems.		
M2.01	Commissioning phases and organization.	1	Understanding
M2.02	Pre-commissioning and post-commissioning documentation.	2	Understanding
M2.03	Process commissioning and associated problems.	2	Applying
M2.04	Plant inspection, non-destructive testing, and leak monitoring.	2	Applying
M2.05	Performance monitoring of plant systems.	2	Understanding
	Series test -1	1	
Contents: Commissioning phases, documentation, and associated problems. Plant inspection: pressure vessels, piping, non-destructive testing. Leak testing and performance monitoring (corrosion, vibration, etc.).			
CO3	Gain practical knowledge in plant operations, including emergency procedures and safety protocols.		
M3.01	Operating discipline and procedure format.	1	Understanding
M3.02	Emergency procedures, handover, and permit system	2	Applying
M3.03	Start-up and shutdown operation of refinery units.	2	Understanding
M3.04	Operating activities, hazards, and exposure of personnel.	2	Applying
M3.05	Corrosion prevention for underground pipes.	2	Understanding

Contents: Operating procedures and safety hazards. Emergency handling and start-up/shutdown protocols. Trip systems, hazards, and corrosion prevention			
CO4	Understand plant maintenance, modification, emergency planning, and storage safety measures for hazardous materials.		
M4.01	Management of maintenance, hazards, and preparation for maintenance.	2	Understanding
M4.02	Isolation, purging, cleaning, and confined space safety.	2	Applying
M4.03	Hot works, tank cleaning, and online repairs.	2	Applying
M4.04	Plant modification, control of modifications, and maintenance of protective devices.	2	Understanding
M4.05	Emergency and disaster planning: onsite and offsite emergency management.	2	Understanding
M4.06	Storage of hazardous materials: LPG, LNG, and other chemicals.	2	Understanding
M4.07	Hazard assessment for LPG, LNG, and other chemical storages.	2	Applying
	SeriesTest–II	1	
Contents: Maintenance management: preparation, safety, and equipment. Hot works, confined space safety, and tank cleaning. Emergency planning, disaster planning, and APELL (Accident Prevention and Emergency Lessons). Storage safety for LPG, LNG, toxic chemicals, and underground storage systems.			

Text / Reference:

T/R	BookTitle/Author
R1	Lees, F.P., “Loss Prevention in Process Industries,” Butterworth Publications, London, 3rd edition, 2005.
R2	Sanoy Banerjee, “Industrial Hazards and Plant Safety,” Taylor & Francis, London,
R3	Fawcett, H., and Wood, “Safety and Accident Prevention in Chemical Operations,” Wiley Interscience, 2nd Edition, 1984.

R4	McElroy, Frank E., "Accident Prevention Manual for Industrial Operations," NSC, Chicago, 1988.
R5	Green, A.E., "High Risk Safety Technology," John Wiley and Sons, 1984.